

Recall $T: V \rightarrow W$ linear map $\Leftrightarrow T(\sum c_i v_i) = \sum c_i T(v_i)$

We saw $\{\text{Linear maps from } \mathbb{R}^p \text{ to } \mathbb{R}^q\} \xleftrightarrow{\text{bijection}} \{q \times p \text{ matrices}\}$
 $(x \mapsto Ax) \longleftrightarrow A$
inverse
bijections
 $\uparrow$
check!
 $T \longmapsto [T(e_1) \ T(e_2) \ \dots \ T(e_p)]$,
where $e_1, e_2, \dots, e_p$ is the standard basis of $\mathbb{R}^p$

Today: matrix of a linear map $V \xrightarrow{T} W$ for any f.d.v.s V, W .
by picking a basis for V and a basis for W

Recall how we get coordinates of a vector w.r.t. a basis

① Suppose $v_1, v_2, \dots, v_n$ is a list of vectors in a v.s.p V
We can define a map $\mathbb{R}^n \xrightarrow{\psi} V$ by

$$\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \xrightarrow{\psi} \sum c_i v_i$$

$$\psi\left(\alpha \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}\right) = \alpha \psi\left(\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}\right)$$

$$\psi\left(\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} + \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}\right) = \psi\left(\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}\right) + \psi\left(\begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}\right)$$

Observe

• ψ is linear

• ψ is onto $\Leftrightarrow v_1, \dots, v_n$ span V

• ψ is injective $\Leftrightarrow v_1, \dots, v_n$ are lin indep

$\left. \begin{array}{l} \psi \text{ is an isomorphism} \\ \Leftrightarrow (v_1, \dots, v_n) \text{ is} \\ \text{a basis for } V \end{array} \right\}$

an ordered

② Take $\hookleftarrow$ basis $B = (v_1, \dots, v_n)$ for n -dimensional v.space V

Then we have the isomorphism $\psi: \mathbb{R}^n \rightarrow V$ as above

Inverse of ψ gives coordinates of any $v \in V$ w.r.t given basis

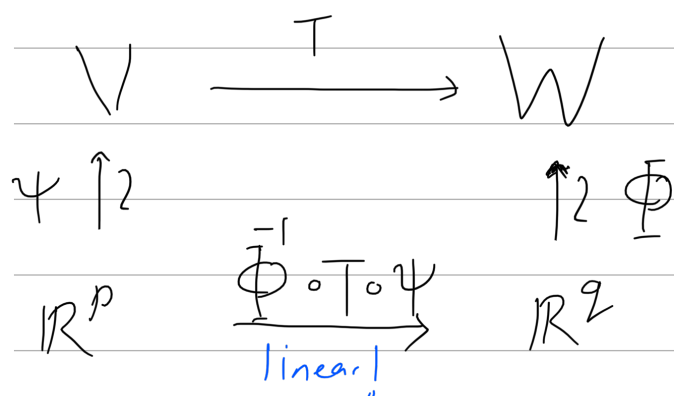
i.e. write $v = \sum_{i=1}^n c_i v_i$ (uniquely b/c v_i 's form a basis)

Then coordinate vector of v wrt. basis B is $\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \Psi(v)$

Back to linear map $V \xrightarrow{T} W$. We will see how to describe T using a matrix after picking bases.

Let $v_1, \dots, v_p$ be an ordered basis for V

Let $w_1, \dots, w_q$ W



$$v \xrightarrow{T} T(v)$$

$$\begin{array}{ccc}
 \downarrow & & \downarrow \\
 \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix} & \xrightarrow[\text{to a linear map from } \mathbb{R}^p \text{ to } \mathbb{R}^q]{T \text{ transported}} & \begin{bmatrix} b_1 \\ \vdots \\ b_q \end{bmatrix}
 \end{array}$$

Write $v = \sum_{i=1}^p c_i v_i$

$\mathbb{R}^p$ to $\mathbb{R}^q$ (namely $\Phi^{-1} \circ T \circ \Psi$)

$$T(v) = \sum_{j=1}^q b_j w_j$$

We must have a matrix A such that

$$A \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_q \end{bmatrix}$$

Question

How to write the matrix A ? Its columns are images of std basis vectors under the linear map $\Phi^{-1} \circ T \circ \Psi$.

$$e_i \xrightarrow{\Psi} v_i \xrightarrow{T} T(v_i) \xrightarrow{\Phi^{-1}} \text{coordinate vector of } T(v_i) \text{ wrt basis } w_1, \dots, w_q$$

Answer. Matrix of T wrt. given bases of V & W is obtained as follows: Its columns are coordinate vectors of $T(\text{given basis of } V)$ wrt given basis of W .

Let's carry this out in example!

① $(V = \text{polynomials of deg} \leq 3) \xrightarrow{T=\text{deriv}} (W = \text{poly of deg} \leq 2)$
wrt. basis $= (1, x, x^2, x^3)$ basis $= (1, x, x^2)$

Matrix of T will have size 3×4 . We build it column by column

$$\text{Answer} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \quad \begin{aligned} T(1) &= 0 = 0 \cdot 1 + 0x + 0x^2 \\ T(x) &= 1 = 1 \cdot 1 + 0 \cdot x + 0 \cdot x^2 \\ T(x^2) &= 2x = 0 \cdot 1 + 2 \cdot x + 0 \cdot x^2 \\ T(x^3) &= 3x^2 = 0 \cdot 1 + 0 \cdot x + 3 \cdot x^2 \end{aligned}$$

If we had chosen $W = \text{poly of deg} \leq 10$ as codomain, we would get the 11×4 matrix obtained by appending 8 rows $0 \ 0 \ 0$ below the above matrix.

You do: same for $V = \text{poly of deg} \leq n$

Important Exercise

② Look at the proof of dimension formula. There we had basis $k_1, \dots, k_r, v_1, \dots, v_r$ for V . We saw that $T(v_1), \dots, T(v_r)$ was a basis for $\text{Im}(T)$. Complete this latter basis to a basis for W . Write matrix of T wrt these bases! (Note: Changing order of one of these bases will make the matrix even nicer.)

Exercises from yesterday. Matrices of given T

(i) $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ rotation by θ around origin. This is linear as seen by geometry. Its matrix is

$T(e_1) = (\cos \theta)e_1 + (\sin \theta)e_2$
 $T(e_2) = (-\sin \theta)e_1 + (\cos \theta)e_2$
 $\cos(\frac{\pi}{2} + \theta)$ $\sin(\frac{\pi}{2} + \theta)$

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

(ii) $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ rotation by θ around z axis

$$\begin{matrix} & e_1 & e_2 & e_3 \\ \begin{matrix} e_1 \\ e_2 \\ e_3 \end{matrix} & \begin{bmatrix} * & & 0 \\ & * & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

(iii) $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ reflection in $y=x$

$$T(e_1) = e_2$$

$$T(e_2) = e_1$$

$$\text{Matrix of } T = \begin{matrix} & e_1 & e_2 \\ \begin{matrix} e_1 \\ e_2 \end{matrix} & \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{matrix}$$

Matrix of T wrt basis e_1+e_2, e_1-e_2 for both $\mathbb{R}^2$'s. ?

Answer

$$\begin{matrix} & e_1+e_2 & e_1-e_2 \\ \begin{matrix} e_1+e_2 \\ e_1-e_2 \end{matrix} & \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \end{matrix}$$